

BAAZ JKBOSE



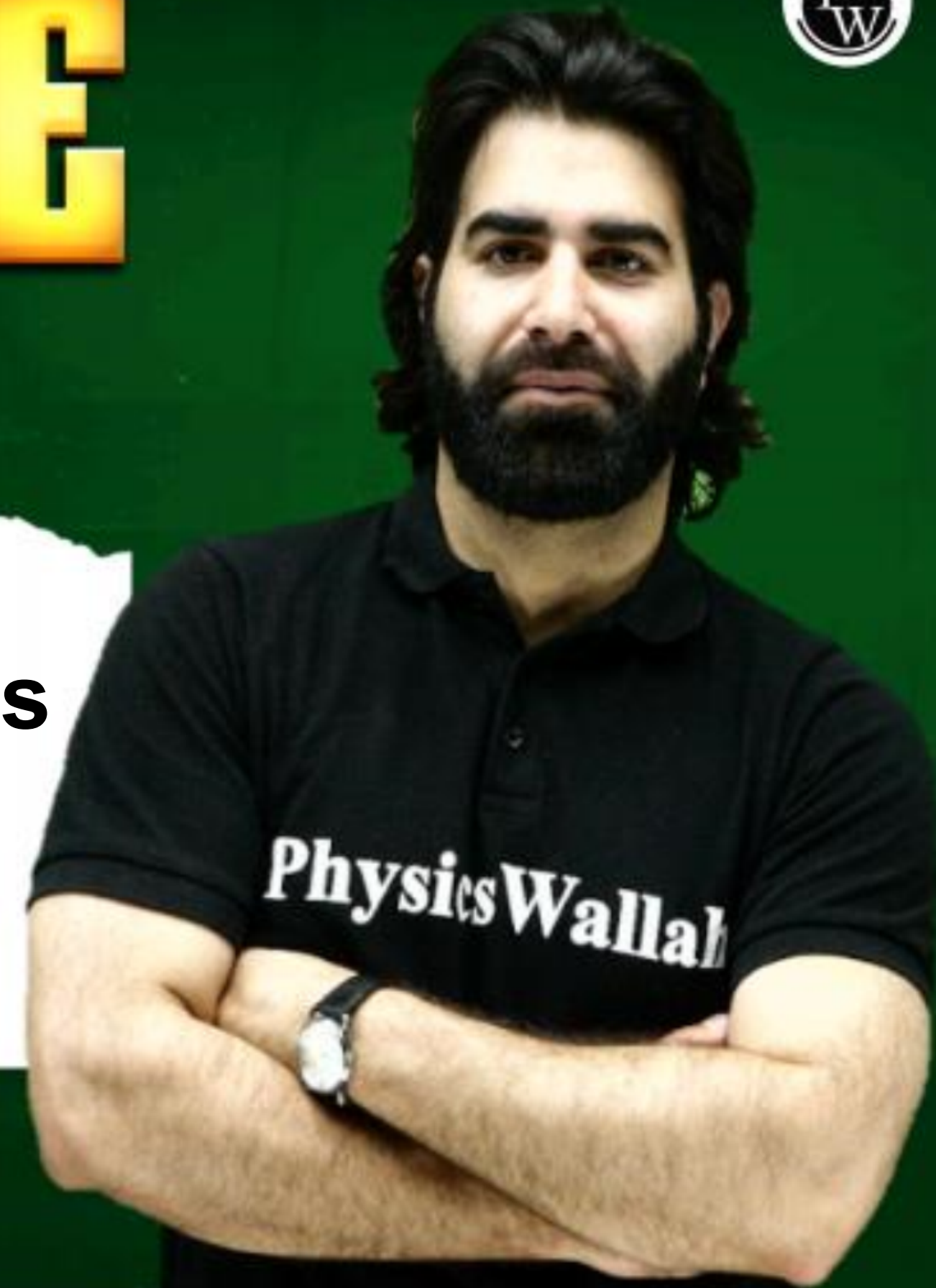
2026 CLASS-11TH

Breathing and Exchange of Gases

LECTURE-4

Zoology

Dr Adil Altaf sir



Topics To Be Covered

- 1 Lung volumes and capacity
- 2

Revision

<u>Mechanism of Breathing</u>	Inspiration	Expiration
External I.C.M.	A. Contraction.	B. Relaxation.
Diaphragm.	c. Contraction.	D. Relaxation.

Lung Volumes

- It is amount of Air Inhaled / Exhaled during various respiratory conditions.
- Device used - Spirometer.
- Test - Spirometry. (Type of Lung Functional Test)

Lung Volumes

① Tidal Volume

→ Amount of Air Breathing or Breathing out during normal Respiration.

$TV = 500ml$ [Normal Respiratory Rate = 12-16 Times/Min].

Total TV in One Minute = $12 \times 500 - 16 \times 500ml/Min$
 $= 6000 - 8000ml/Min$

② Inspiratory Reserve Volume

→ Additional/Extra Amount of Air a person can inspire (Forefully), after normal Inspiration.

$IRV = 2500ml - 3000ml$

③ Expiratory Reserve Volume

- Additional/Extra Amount of Air a person can expire (Forefully), after normal Expiration.

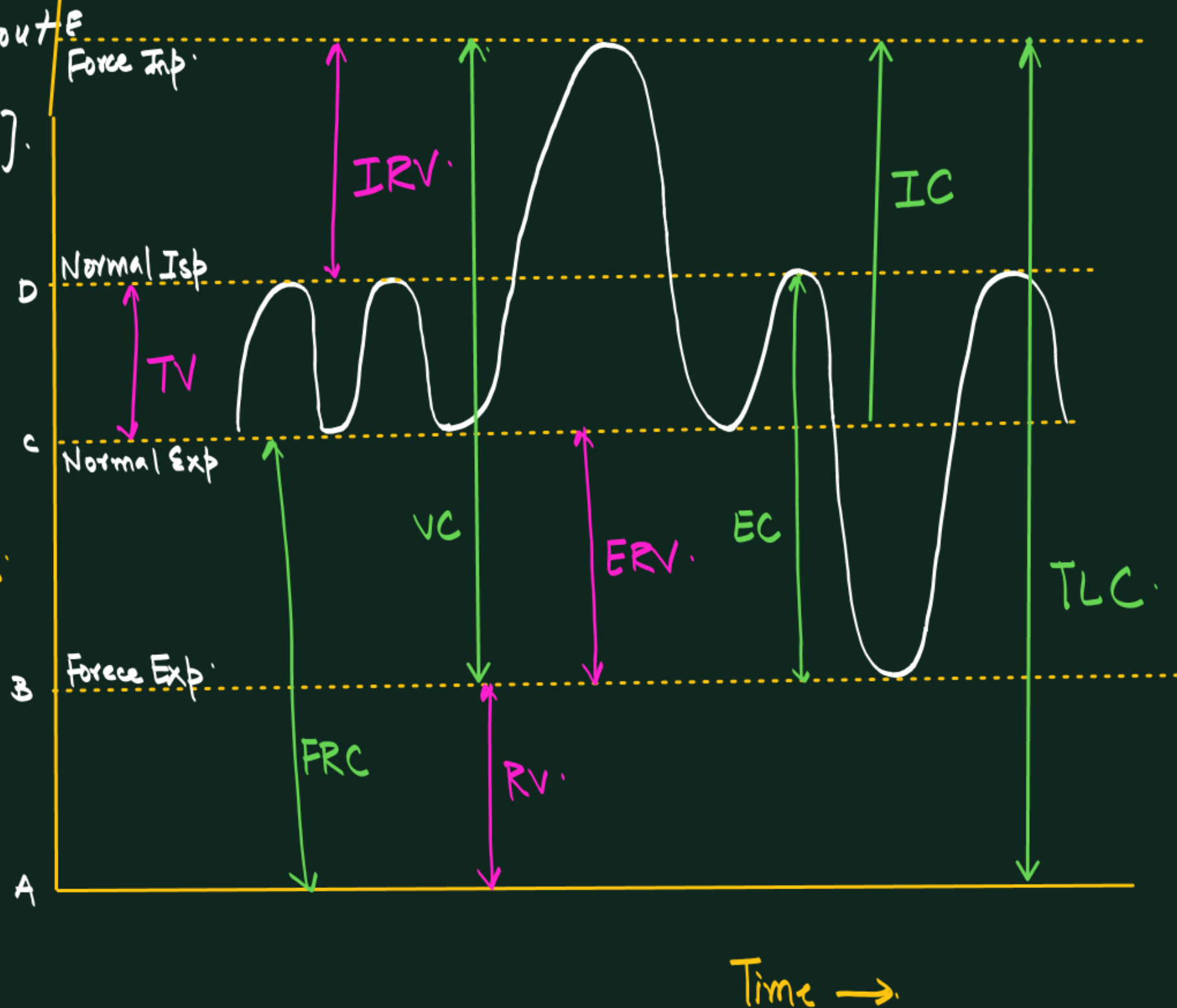
$ERV = 1000ml - 1100ml$

④ Residual Volume

→ Amount of Air which Remains in Lungs even after Foreful Expiration.
 [Can't be Measured using Spirometre].

$RV = 1100 - 1200ml$

Amount of Air In Lungs



Lung Capacities → Combination of Lung Volumes.

① **Inspiratory Capacity** → Maximum Amount of Air one can Inspire After Normal Expiration.
 $IC = TV + IRV$

② **Expiratory Capacity** → Maximum Amount of Air one can Expire After Normal Inspiration.
 $EC = TV + ERV$

③ **Vital Capacity** → Maximum Amount of Air one can expire After Forceful Inspiration.
or
→ Maximum Amount of Air one can Inspire After Forceful Expiration.
$$\left[\begin{array}{l} VC = \overline{ERV + TV + IRV} \\ VC = \overline{EC} + IRV \\ VC = ERV + \overline{IC} \end{array} \right]$$

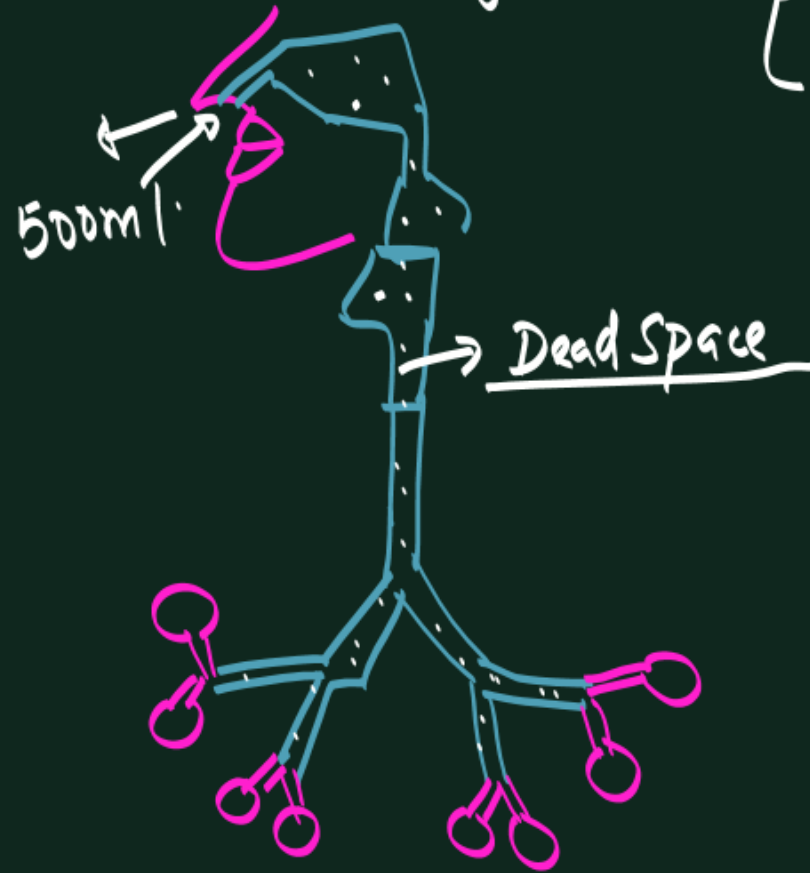
④ **Functional Residual Capacity** → Amount of Air Remaining in Lungs After Normal Expiration.
 $FRC = ERV + RV$

⑤ **TLC = Total Lung Capacity**
 $TLC = \overline{IRV + TV + ERV} + RV = VC + RV$
 $TLC = IC + FRC$

Dead Space → Amount of Air which is not Available for gaseous Exchange.

→ Because it is present in Conduction Zone

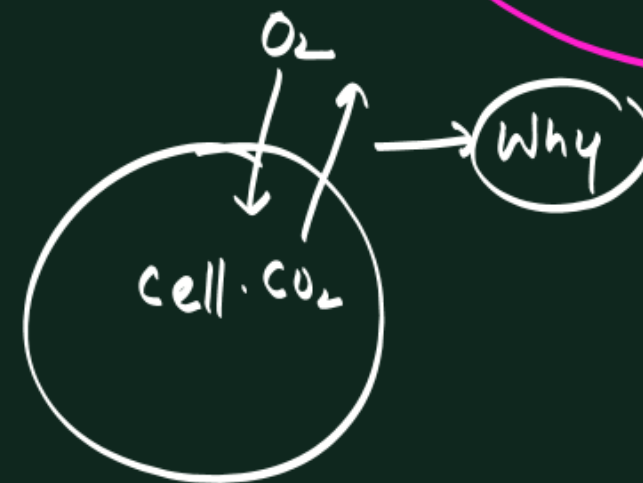
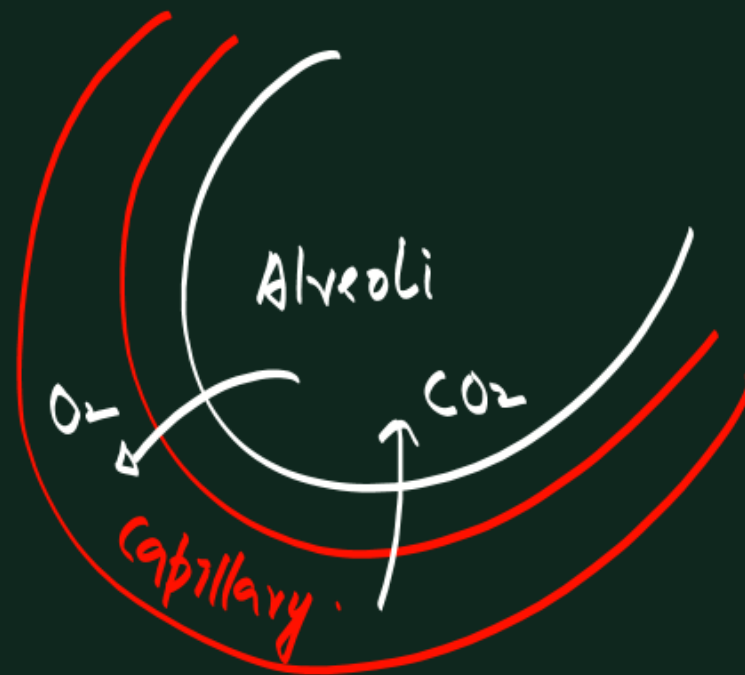
→ It is About 150ml of TV. $\left[\begin{array}{l} \text{Effective} \\ \text{Amount} \end{array} = 500 - 150 \text{ ml} \right]$
 $= 350 \text{ ml}$



→ Gaseous Exchange / Exchange of Gases

↳ Exchange of Gases through Diffusion Membrane, (Thickness < 1mm) occurs due to difference in their partial pressures.

↳ Gases Move From their Higher partial pressure towards lower partial pressure.



- Mole Fraction

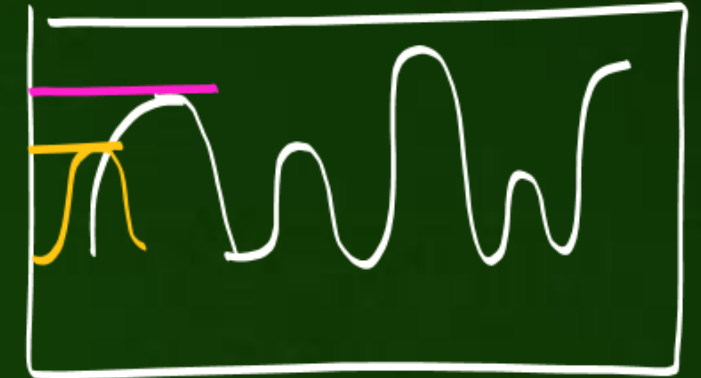
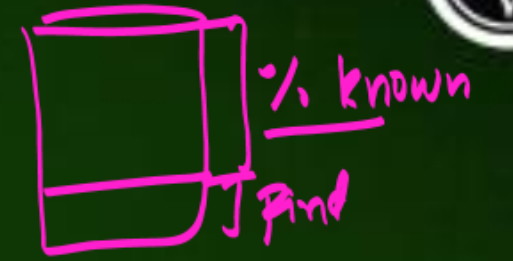
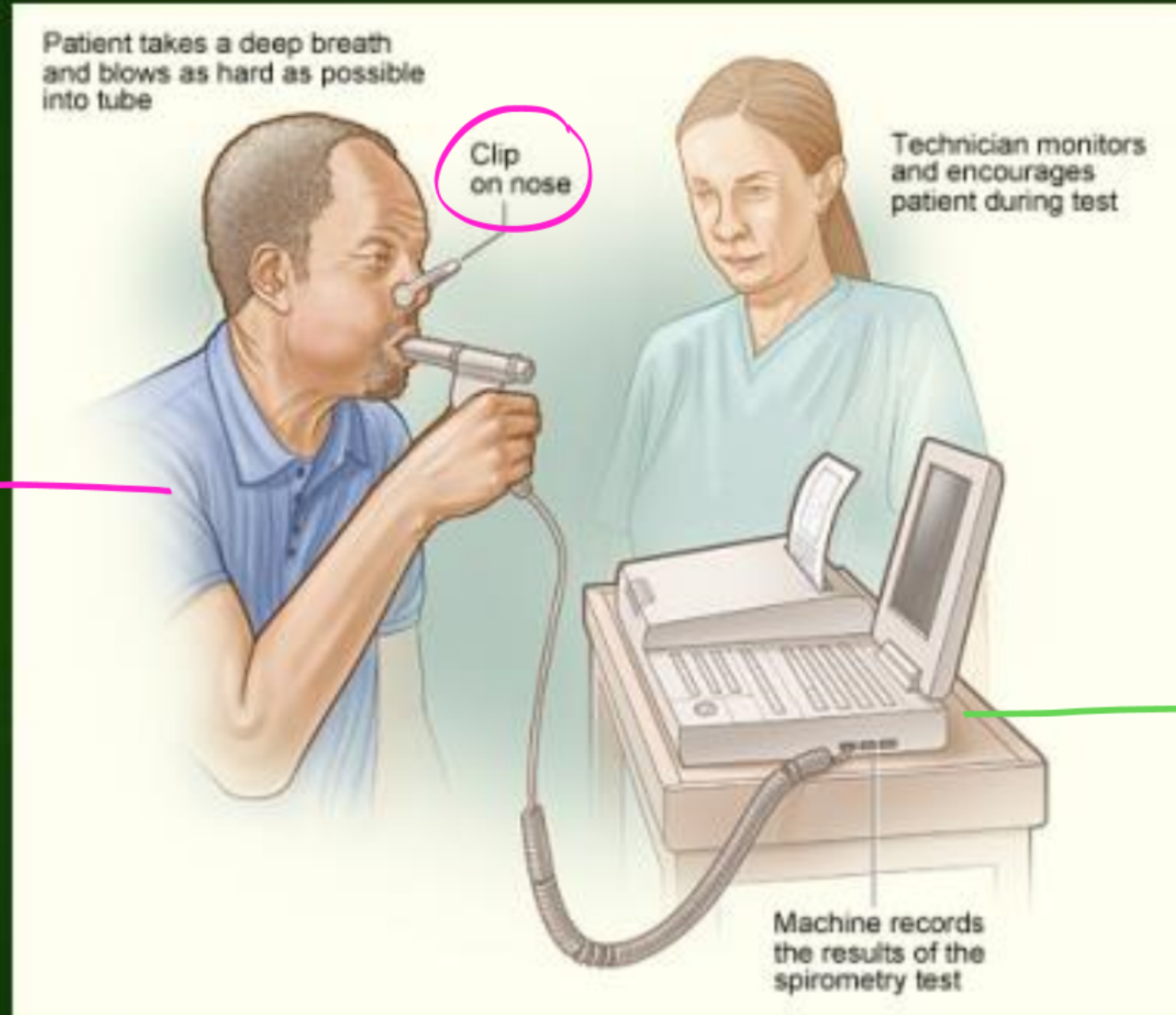
- Dalton's Law (gases)

$$\hookrightarrow P_T = P_1 + P_2 + P_3 - \dots$$

	Atmosphere	Alveoli	Deoxygenated Blood	oxygenated Blood	Tissue.
PO_2	159	104 	40	95 	40
PCO_2	0.3	40 	45	40 	45

Atmosphere
 P_{N_2} P_{O_2}
 P_{CO_2}

$$P_T = P_{N_2} + P_{O_2} + P_{CO_2} + \dots$$



(Graph Represent)

Spirometre

— Volume of Air Inhaled/Exhaled

QUESTIONS

Which of the following constitute the conduction part of the respiratory system.

- 1 External nostrils to trachea
- 2 External nostrils to primary bronchioles
- 3 Respiratory bronchioles to alveoli
- 4 External nostrils to terminal bronchioles

QUESTIONS

Which of the following events happen when air enters lungs.

Inspiration .

- 1 Ribs and sternum raised
- 2 Volume of thorax increased
- 3 Diaphragm contracted
- 4 All of the above

Thank
You